Screening Financial Disclosure Risk

Textual–Financial Feature Fusion and Explainable Anomaly Detection

Evidence from Chinese A-share Manufacturing Firms, 2021–2023

Group 12A2 · Intro FinTech

December 2025

DATA
CSMAR statements
+ CNINFO reports

FEATURES
6 financial
3 MD&A text

RANKING
Isolation Forest
+ SHAP

MOTIVATION

Three research questions

01 **Does MD&A text add detection value beyond financial features?**
Fused versus finance-only ranking against the same benchmark.

02 **Which feature group drives the ranking?**
Global and local SHAP attribution on the fused model.

03 **Is the ranking stable across models and thresholds?**
Autoencoder robustness, threshold sensitivity, list overlap.

SAMPLE CONSTRUCTION

Data: statements, reports, one benchmark

STRUCTURED DATA

CSMAR 2020–2023

Six statement ratios

Balance-sheet and income items

DISCLOSURE TEXT

CNINFO 2021–2023

Annual report PDFs

MD&A section located per report

EVALUATION BENCHMARK

Beneish M-Score

Evaluation only

Never used in training

SAMPLE FRAME

3,694 firms · 10,172 firm-years

Complete features: 8,140 firm-years (80.0%) across 3,265 firms

PDF coverage 98.8% of the frame

MD&A usable text 96.7%–99.1% by year

Missing files are pre-listing NEEQ years

Frame 10,172 firm-years, usable sample 8,140.

Balanced across 2021 (2,430), 2022 (2,769) and 2023 (2,941).

METHOD OVERVIEW

Pipeline: five reproducible steps

STEP 1 **Ingest** → STEP 2 **Extract MD&A** → STEP 3 **Construct 9 features** → STEP 4 **Rank anomalies** → STEP 5 **Evaluate & explain**

Versioned inputs

Raw CSMAR workbooks and CNINFO PDFs are hashed and logged.

Checkpointed extraction

MD&A extraction writes after every 500 reports.

Single command

The whole chain runs from one entry script.

No benchmark variable enters model training.

FEATURE SYSTEM

Nine features, two groups

6 FINANCIAL FEATURES

- roa
- leverage
- revenue_growth
- ocf_to_assets
- receivables_turnover
- accruals_to_assets

3 MD&A TEXT FEATURES

- **readability**
  characters per sentence proxy
- **sentiment_balance**
  (pos − neg) / (pos + neg)
- **uncertainty_ratio**
  uncertainty words / tokens

Winsorise 1% / 99% · subindustry-year z-score · token-level lexicon match

MODEL DESIGN

Unsupervised screening, four rankings

Isolation Forest
primary ranking model

Finance-only	Text-only	Fused	Autoencoder 16–8–16

500	256	42	10%
trees	max samples	seed	review queue

Identical features, sample and evaluation for every ranking.

Scores are review-priority signals, not fraud verdicts.

MAIN RESULT

RQ1: text adds no detection value

Spearman rank agreement with the Beneish M-Score

Finance-only
0.136
rho, n = 8,140

Text-only
-0.029
rho, n = 8,140

Fused
0.103
rho, n = 8,140

AUTOENCODER
0.009
not significant (p = 0.41)

Enrichment under the literal −1.78 cut-off

Review queue	Finance-only	Fused
top 5%	1.370	1.346
top 10%	1.217	1.198
top 15%	1.168	1.135

Recalibrated (sample top decile)

Review queue	Finance-only	Fused
top 5%	2.555	2.727
top 10%	1.867	1.843
top 15%	1.638	1.532

Answer: no. Adding the three MD&A features does not improve the ranking (0.103 vs 0.136); the text-only ranking is essentially uncorrelated (rho = -0.029).

The negative result holds under both benchmark definitions.

EXPLAINABILITY

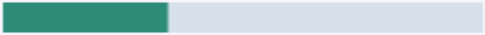
RQ2: financial features dominate

Global SHAP contribution

Financial group 65.4%



Textual group 34.6%



Three most important global features

1. ocf_to_assets 0.322
2. revenue_growth 0.320
3. readability 0.317

LOCAL EXPLANATION

Top-ranked firm-year 000820 (2021)

Subindustry C42 · anomaly score 0.759

Main local drivers

- receivables_turnover
- revenue_growth
- accruals_to_assets

M-Score 1,077.5 makes this an extreme observation, not a confirmed case of misconduct.

Screening signals support review; they do not prove misconduct.

ROBUSTNESS

RQ3: stable lists, fragile ranking

Top-10% list overlap (Jaccard)

	Fin	Txt	Fused	AE
Fin	—	0.111	0.817	0.373
Txt	0.111	—	0.256	0.147
Fused	0.817	0.256	—	0.362
AE	0.373	0.147	0.362	—

AUTOENCODER ROBUSTNESS

rho = 0.009

The fused autoencoder does not reproduce the Isolation Forest ranking, so the headline result is model-dependent.

Finance–fused overlap 0.817
Finance–text overlap 0.111

Threshold sensitivity

Queue	Legacy	Recalibrated
5%	1.346	2.727
10%	1.198	1.843
15%	1.135	1.532

Fused-model enrichment at 5%, 10% and 15% review queues; the ranking order does not change.

INTERPRETATION

Limits we report ourselves

Dictionary sentiment is a proxy

A published financial lexicon matched token by token; nuance and context are lost.

Extraction coverage is incomplete

Scanned or unreadable reports are excluded; usable text is 96.7%–99.1% by year.

M-Score is an imperfect benchmark

The literal cut-off flags 62.8% of this sample, so we also report a sample-calibrated decile.

Outputs are review-priority signals, not misconduct labels.

No fraud, violation or disclosure-quality verdict is claimed anywhere in the paper.

OPEN SCIENCE

Reproducible by construction

- **Seed fixed**
  SEED = 42 in every stochastic step
- **Versioned scripts**
  Numbered run_0x scripts, one job each
- **Versioned intermediates**
  Every CSV table in outputs/tables is released
- **Checkpointed extraction**
  Downloads and MD&A parsing resume after interruption
- **One command**
  run_99_full_pipeline.py rebuilds every table and figure

FULL REBUILD

python scripts/
run_99_full_pipeline.py

Deterministic output, documented in the reproducibility guide.

Manuscript §5 is generated from the same tables by a single assembly script, so numbers in the paper cannot drift from the released results.

TAKEAWAYS

Conclusion

Financial features carry a weak but significant screening signal (rho = 0.136); MD&A text features add none.

What we deliver
A reproducible, explainable screening pipeline built on public data.

What we find
Text-only and fused rankings fail to improve on finance-only.

What it means
Textual disclosure risk is not captured by simple lexicon measures here.

What is next
Inquiry letters, restatements and audit opinions as outcome benchmarks.

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Thank you · Questions